

Shu Wen

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PROFESSIONAL SUMMARY

Versatile mechanical engineering student specializing in solid mechanics & design and mechatronics, with over 3 years of hands-on experience in designing and analyzing mechanical components and systems. Interested in robotics, automotive parts, and intelligent systems, and real-world applications.

Technical Skills: SolidWorks, AutoCAD, PDM, MBD, Autodesk, Python, ANSYS, COMSOL, LTSpice, MATLAB, Fusion 360, Minitab

Languages: English, Mandarin Chinese

WORK EXPERIENCE

Thermal and Mechanical Engineer

May 2025 – May 2026

Litens Automotive Group

Ontario, Canada

- Designed and optimized thermally relevant EV mechanical components and assemblies in SolidWorks within an R&D project, focus heavily on 3D printing and rapid prototyping
- Researched, designed, assembled, and validated a 70:1 humanoid cycloidal actuator, conducting rigorous torque-speed characterization and market benchmarking to optimize performance against industry standards.
- Developed and validated 2D drawings with GD&T and tolerance specifications, integrating high-precision measurement data into 3D models for reverse-engineering design and tolerance stack-up evaluation.
- Developed a versatile, single-plate test rig capable of switching between high-precision 3 Nm deadweight calibration and 40 Nm high-load electrical characterization for gear train validation.
- Performed FEA simulations in SolidWorks Simulation to evaluate multi-gear behavior under varying torque and load conditions. Leveraged material research and simulated stress concentration results to recommend gear materials, enhancing performance, durability, and reliability.

Structural Engineer

June 2024 – August 2024

Yuwell Medical Devices

Nanjing, China

- Modeled and designed vibration-damping and ventilation components for oxygen concentrator compressors using SolidWorks, optimizing system performance and reliability.
- Utilized 3D printing to prototype the designed module and conducted rigorous performance tests on different points of the module, successfully maintaining the overall compressor temperature below 55 Celsius degrees for continuous operation over 1.5 hours.
- Acquired hands-on expertise with oxygen concentrators through research-focused projects, emphasizing the enhancement of component longevity, thermal efficiency, heat management, and manufacturability to ensure durability and reliability in high-performance medical applications.

Undergraduate Research Assistant

May 2024 - June 2024

NUS Multiphase Oil-Water-Air Flow Loop Laboratory, National University of Singapore

Singapore

- Investigated the distinct flow behavior of gasoline and air in a high-velocity pipeline environment by conducting CFD simulations and predictive analyses using ANSYS and COMSOL software to model the movement and interaction of multiple media within pipelines.
- A wavy annular flow is successfully obtained by using the VOF method and FEA analysis is also conducted based on the pipe to analyze the sustainability and stability of the pipe.
- Authored a comprehensive research paper (12+ pages) discussing the simulation outcomes and how the predictions can be applied to improve the safety and efficiency of two-phase fluid transportation.

EDUCATION

University of Toronto

Sep 2022 – present

Bachelor of Applied Science in Mechanical Engineering + PEY Co-op

Toronto, ON

Certifications:

- CSWA Certificate
- George Brown Machining Certificate